

14-DARS

While tsikli

Shartli takrorlash, break, continue va cheksiz tsikllar

1. While nima?

while tsikli sharti **True** bo'lguncha kodni takrorlaydi. **for** aniq son marta ishlasa, **while** shart bajarilguncha ishlaydi:

```
i = 1
while i <= 6:
    print(i)
    i = i + 1
```

1 dan 6 gacha chiqadi.

Eslatma: Tsikl ichida shartni o'zgartirishni unutmang ($i = i + 1$). Aks holda tsikl cheksiz aylanadi!

2. break — tsiklni to'xtatish

break tsiklni darhol to'xtatadi:

```
while True:
    son = input('Son kiriting (chiqish: x): ')
    if son == 'x':
        break
    print(f"Siz {son} kiritdingiz")
```

x kiritilsa tsikl tugaydi.

3. continue — keyingi aylanaga o'tish

continue joriy aylanani tashlab, keyingisiga o'tadi:

```
i = 0
while i < 6:
    i += 1
    if i == 3:
        continue # 3 ni tashlab ketadi
    print(i)
```

3 dan tashqari hammasi chiqadi.

4. Amaliy misol: yig'indi

```
jami = 0
son = int(input('Son (0 - to\'xtatish): '))
while son != 0:
    jami += son
    son = int(input('Son: '))
print(f'Yig'indi: {jami}')
```

0 kiritilguncha sonlarni qo'shadi.

5. while ... else

```
i = 1
while i <= 3:
    print(i)
    i += 1
else:
    print('Tsikl tugadi')
```

else tsikl normal tugaganda ishlaydi.

Xulosa

- **while** shart rost bo'lguncha takrorlaydi.
- Shartni o'zgartiring, aks holda cheksiz tsikl.
- **break** to'xtatadi, **continue** keyingiga o'tadi.